COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3363	ARTIFICIAL INTELLIGENCE FOR BUSINESS	30	0	30	30	90	3

LEARNING OUTCOMES

CO1	Understand and explain the fundamental concepts, branches, and applications of Artificial Intelligence across various business functions.
CO2	Apply machine learning algorithms such as Linear Regression, KNN, Decision Trees, and Random Forest to solve business problems using appropriate tools.
CO3	Analyze and implement clustering methods and recommendation systems for personalized business applications using relevant techniques.
CO4	Build and evaluate deep learning models using neural network architectures and frameworks like TensorFlow and PyTorch for business-specific applications.
CO5	Evaluate and implement generative AI applications using techniques like GANs and LLMs, considering ethical implications and business impact.

UNIT 1 INTRODUCTION TO ARTIFICIAL INTELLIGENCE 12

Basic Concepts of Artificial Intelligence – Branches of AI – Machine Learning – Deep Learning – Natural language Processing – Robotics Expert Systems – Vision – Benefits and Risks of AI – Applications of AI – Benefits and Risks of AI – Application of AI in Business – AI in Marketing, Finance, Operations and HR.

UNIT 2 MACHINE LEARNING 12

Overview of Machine Learning – Types of Machine Learning – Supervised Learning – Unsupervised Learning – Reinforcement Learning – Machine Learning Framework – Challenges of ML – Role of Statistics in ML - Learning Algorithms and its Applications in Business – Linear Regression – KNN– Decision Trees - Ensemble Learning – Random Forest Build ML applications using relevant tools and techniques.

UNIT 3 CLUSTERING AND RECOMMENDATION ENGINES 12

Clustering Techniques – K-Means – Clustering Applications in Business – Recommendation Engine – Recommendation System Techniques – Content Based Recommendations – Collaborative Recommendations – Applications of Recommendation Engines in Business- Build ML applications using relevant tools and techniques.

UNIT 4 DEEP LEARNING **12**

Overview of Deep Learning – Difference between ML and Deep Learning – Neural Networks and their Architectures – Feedforward Neural Networks – Convolutional Neural Networks (CNNs) – Recurrent Neural Networks (RNNs) – Transformers – Deep Learning for Computer Vision (Image Classification, Object Detection, Image Segmentation) – Deep Learning for Speech Learning (Speech Recognition, Text-to-Speech, Speaker Identification) – Deep Learning Frameworks (TensorFlow, PyTorch) – Optimization Techniques – Hyperparameter Tuning – Challenges in Deep Learning – Applications of Deep Learning in Business – Implementing Deep Learning Models using relevant tools and techniques.

UNIT 5 GENERATIVE AI **12**

Overview of Generative AI – Generative vs Discriminative Models – Tokenization Embeddings - Autoencoders – Generative Adversarial Networks (GANs) - Large Language Models (LLMs) – Prompt Engineering – Fine-Tuning Pre-trained Models – Ethical Considerations in Generative AI – Role of Generative AI in Business and Creativity – Case Studies of AI-Generated Content – Implementing Generative AI Applications using relevant tools and techniques.

TOTAL 60 HOURS

REFERENCE BOOKS

1. Rajendra Akerkar, Artificial Intelligence for Business, Springer, 2019.
2. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th edition, Pearson Publications, 2020.
3. Aurélien Géron, Hands on Machine Learning with Scikit-Learn, Keras and TensorFlow: Concepts, Tools and Techniques to Build Intelligent Systems, O'Reilly, 2022.
4. Andreas C. Müller & Sarah Guido, Introduction to Machine Learning with Python: A Guide for Data Scientists, O'Reilly, 2016.
5. Steven Finlay, Artificial Intelligence and Machine Learning for Business: A No-Nonsense Guide to Data Driven Technologies, 3rd Edition, Relativistic, 2019.
6. Sandeep Kumar Panda, Vaibhav Mishra, R. Balamurali and Ahmed A.Elngar, Artificial Intelligence and Machine Learning in Business Management: Concepts, Challenges and Case Studies, CRC Press, 2024.

CO PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2				2
CO2	3	3	3	2				2
CO3	3	3	3	2				2
CO4	3	3	3	2				2
CO5	3	3	3	2				2

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3353	FUNCTIONAL ANALYTICS	45	0	0	45	90	3

LEARNING OUTCOMES

CO1	Apply appropriate analytical methods to solve business problems in Marketing.
CO2	Apply appropriate analytical methods to solve business problems in Finance
CO3	Apply appropriate analytical methods to solve business problems in Operations
CO4	Apply appropriate analytical methods to solve business problems in HR
CO5	Demonstrate knowledge and critical understanding of analytics in different domains

UNIT 1 ANALYTICS IN MARKETING **9**

Basics of Marketing Management - Key Performance Indicators/Drivers -Baseline and Benchmark KPI's - KPI Tree for Marketing - Marketing Dashboards - Application of Analytics in Marketing- Applications of AI and Gen AI in marketing.

UNIT 2 ANALYTICS IN FINANCE **9**

Basics of Financial Management - Key Performance Indicators/Drivers of Finance function, Baseline and Benchmark KPI's -KPI Tree for Finance - Finance Dashboards - Application of Analytics in Finance- Applications of AI and Gen AI in Finance.

UNIT 3 ANALYTICS IN OPERATIONS **9**

Basics of Operations Management - Inventory Control - Logistics and Supply Chain - KPI Tree for Operations - Operations Dashboard - Using analytics in Manufacturing and Service Operations - Preventive Maintenance in Manufacturing – Forecasting and Supply Chain Analytics- Applications of AI and Gen AI in Operations.

UNIT 4 ANALYTICS IN HR **8**

Basics of HR Management - Key Performance Indicators/Drivers of HR function, Baseline and Benchmark KPI's - KPI Tree for HR - HR Dashboards- Application of Analytics in HR- Applications of AI and Gen AI in HR

UNIT 5 APPLICATION OF ANALYTICS IN DOMAINS **10**

Use of Analytics in different domains such as Healthcare, Entertainment, Telecommunication, IT /ITES, Hospitality, E-Commerce, Retail, Manufacturing, Banking and Financial Services Industry.

TOTAL 45 HOURS

REFERENCE BOOKS

1. Marketing Analytics: A Practical Guide to Improving Consumer Insights Using Data Techniques, Mike Grigsby,Kogan Page, 2nd Edition,2018.
2. Ramashankar Yadav, Sunil Maheswari , HR Analytics: Connecting Data and Theory, Wiley 2020.
3. R.K.Arora, Financial Risk Analytics: Measurement, Management and Examples in R, Wiley, 2022.
4. Pitabas Mohanty, Financial Analytics, Wiley, 2023.
5. Gerardus Blokdyk, Operational Analytics, Second edition, 5STAR Cooks, 2019.

CO PO MAPPING

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	3	3	2				2
CO2	3	3	3	2				2
CO3	3	3	3	2				2
CO4	3	3	3	2				2
CO5	3	3	3	2				2

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3354	DIGITAL TRANSFORMATION FOR ORGANIZATIONAL GROWTH	45	0	0	45	90	3

LEARNING OUTCOMES

CO1	Apply the principles of digital transformation concepts to business scenarios.
CO2	Analyze case studies to understand how organizations have used digital transformation for non-linear growth.
CO3	Integrate various digital technologies to devise a comprehensive solution for driving non-linear growth
CO4	Critically assess the potential impact of proposed digital transformation strategies

UNIT 1 INTRODUCTION TO DIGITAL TRANSFORMATION **8**

Digital Transformation – Role of Digital Transformation in Business Growth – Digital Implementation Methodology – Emerging Technologies – Emerging Digital Themes by Industry.

UNIT 2 DIGITAL TRANSFORMATION METHODS **8**

Digital Transformation Lifecycle – Design Thinking – Clayton Christensen's Disruptive Innovation Framework – Social Technologies – Applying Digital Matrix – Cloud Computing – Collision at the Core, Reinvention at the Root.

UNIT 3 INTELLIGENT TECHNOLOGIES **9**

Robotic Process Automation – Internet of Things – Edge Computing – Artificial Intelligence – Potential AI opportunities – Blockchain – Drones – Virtual and Augmented Reality – 3D Printing – Use Cases.

UNIT 4 APPLICATIONS OF DIGITAL TECHNOLOGIES IN BUSINESS **9**

Creating Data savvy organization using Analytics – Driving Intelligent Automation – Case Studies of Successful Digital Transformation – Driving Non-Linear Growth using Digital Technologies.

UNIT 5 DIGITAL TRANSFORMATION STRATEGIES AND ROADMAP 8

Identifying opportunities for Digital Transformation – Devising a strategy & roadmap for implementing digital technologies.

TOTAL 45 HOURS

REFERENCE BOOKS

1. Anup Maheshwari, Digital Transformation: Building Intelligent Enterprises, Wiley, 2019.
2. David L. Rogers, The Digital Transformation Roadmap, Columbia University Press, 2023.
3. Venkat Venkatraman, The Digital Matrix: New Rules for Business Transformation Through Technology, 1st Edition, Penguin, 2017.
4. Raktim Sing, Driving Digital Transformation: Reshape the Future of Your Business, 9th Edition, Pendown Press, 2020.
5. Siu Loon Hoe, Digital Transformation: Strategy, Execution and Technology, CRC Press, 2022.

CO PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	2	2	1		3
CO2	3	3	2	2	2	1		3
CO3	3	3	2	2	2	1		3
CO4	3	3	2	2	2	1		3

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3364	**TOOLS FOR BUSINESS ANALYTICS**	**30**	**0**	**30**	**30**	**90**	**3**

LEARNING OUTCOMES

CO1	Apply business analytics tools effectively in real-world scenarios.
CO2	Apply core Python libraries for data manipulation and analysis.
CO3	Apply Exploratory Data Analysis (EDA) techniques to prepare data for analysis.
CO4	Evaluate and justify the creation of effective data visualizations and dashboards.
CO5	Evaluate the performance and effectiveness of machine learning models for predictive analytics.

UNIT 1 OVERVIEW OF BUSINESS ANALYTICS TOOLS 8

Introduction to Business Analytics Tools – Categories of Analytics Tools – Comparative Analysis – Strengths and Limitations. Getting Started with Python – Installation of Python and Setting Up the Development Environment – Introduction to Python Environment – Defining Functions – Operators in Python – Data Types and Their Applications – Essential Data Structures – Conditional Execution – Iterative Statements.

UNIT 2 CORE PYTHON LIBRARIES FOR BUSINESS ANALYTICS 10

Overview of NumPy – Fundamentals of pandas – Working with Series and Data frames – Indexing and Reindexing Techniques – Data Selection and Sub setting – Filtering and Sorting Data – Identifying Unique Values – Counting Value Occurrences – Data Import and Storage Options – Introduction to Data Frames Optimization.

UNIT 3 EXPLORATORY DATA ANALYSIS USING PYTHON 12

Introduction to EDA – Importance and Process of EDA – Data Cleaning Techniques – Handling Missig Data – Methods for Replacing Values – Deduplication Strategies – Outlier Detection and Treatment – Feature Scaling and Normalization – Encoding Categorical Variables – Data Transformation – Data Merging and Joining – Implementing Pivot Tables – Grouping and Aggregating Data – Introduction to Feature Engineering.

UNIT 4 DATA VISUALIZATION USING PYTHON 10

Introduction to Data Visualization – Types of Visualizations and Their Applications – Overview of Visualization Libraries – Introduction to Matplotlib – Understanding and Implementing Plotting Functions – Creating Visualizations with Seaborn – Box Plots and Histograms – Count Plots and Pie Charts – Violin and Line Plots – Scatter Plots and Pairwise Analysis – Facet Grids for Multi- Plot Visualizations – Creating Heatmaps – Interactive Visualizations using Plotly – Building Dashboards for Business Insights.

UNIT 5 MACHINE LEARNING MODELING WITH PYTHON 20

Introduction to SciPy for Statistical Computing – Understanding Clustering Techniques – Overview of stats models for Statistical Analysis – Implementing Linear Regression – Introduction to scikit-learn Library – Logistic Regression and Its Applications – Evaluating Model Performance – Introduction to Decision Trees – Cross-Validation Techniques for Model Assessment

TOTAL 60 HOURS

REFERENCE BOOKS

1. Umesh R. Hodeghatta and Umesha Nayak, Practical Business Analytics Using R and Python: Solve Business Problems Using a Data-driven Approach Apress,2023.
2. Galit Shmueli; Peter C. Bruce; Peter Gedeck; Nitin R. Patel; O.P. Wali, Data Mining for Business Analytics, (An Indian Adaptation): Concepts, Techniques and Applications in Python, Wiley, 2023.
3. George Snypes, Python Data Wrangling for Business Analytics, Fiel LLC, 2024.
4. Bharti Motwani, Data Analytics using Python, Wiley, 2020.

CO PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	2				
CO2	3	3	2	2				
CO3	3	3	2	2				
CO4	3	3	3	2				2
CO5	3	3	3	2				2

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3355	BUSINESS INTELLIGENCE THROUGH DATA ENGINEERING	45	0	0	45	90	3

LEARNING OUTCOMES

CO1	Explain the fundamental concepts of Business Intelligence, enterprise data strategy, and data governance.
CO2	Describe the role of data engineering, ETL/ELT processes, and data integration techniques in business intelligence.
CO3	Analyze different data storage architectures, including data warehouses, data lakes, and cloud-based platforms, and their impact on business intelligence.
CO4	Evaluate the challenges of legacy data systems and assess strategies for data modernization and real-time data processing in business transformation.
CO5	Examine case studies of data-driven enterprises to assess the impact of data strategy, AI, and machine learning on business intelligence and decision-making.

UNIT 1 FOUNDATIONS OF BUSINESS INTELLIGENCE & ENTERPRISE DATA STRATEGY 8

DIKW Pyramid – Types of Data – Why Data is the New Oil: The Role of BI in Business – Evolution of Business Intelligence: From Legacy BI to Modern BI – Key Components of Enterprise Data Strategy – Data Governance & Compliance (GDPR, Data Security, and Ethics).

UNIT 2 DATA ENGINEERING & INTEGRATION FOR BUSINESS INTELLIGENCE 8

What is Data Engineering? (And Why Business Leaders Should Care) – ETL vs. ELT: The Modern Approach to Data Integration – ETL Pipelines & Data Workflow Automation – Data Integration & APIs: Connecting Different Business Systems – Enterprise Data Architecture.

UNIT 3 DATA STORAGE & CLOUD SOLUTIONS FOR BI 10

Data Warehouse vs. Data Lake vs. Lakehouse – Data Marts and Their Role in BI -Use Cases of Data Lakes in Business (Marketing, Finance, Supply Chain) – The Rise of the Lakehouse Architecture – Cloud Data Platforms: AWS, Azure, Google Cloud – Traditional Data Storage vs. Modern Data Platforms.

UNIT 4 DATA MODERNIZATION & BUSINESS TRANSFORMATION 9

Challenges of Legacy Data Systems – Cloud-Based vs. On-Premises Data Management – The Shift from Batch Processing to Real-Time Data – Understanding Data Maturity Levels (Reactive → Predictive → Prescriptive) – How Companies Scale Their Data Capabilities.

UNIT 5 DATA STRATEGY, AI, & THE FUTURE OF BUSINESS INTELLIGENCE 10

Case Studies of Data-Driven Enterprises (Amazon, Netflix, Unilever) – Building a Data Culture in Organizations – Aligning Data Strategy with Business Objectives – Data Monetization: How Companies Generate Revenue from Data – Role of AI & Machine Learning in Business Intelligence – The Future: Automated Decision-Making & Data-Driven Leadership.

TOTAL 45 HOURS

REFERENCE BOOKS

1. Joe Reis and Matt Housley, Business Intelligence: The Ultimate Guide to BI, Artificial Intelligence, Machine Learning, Big Data, Cybersecurity, Data Science, and Predictive Analytics , June 2022, O REilly Media, Inc.
2. Dr. Naveen Kumar Singh and Shanu Gupta Business Intelligence and Analytics, September 2024 by Dr. Naveen Kumar Singh (Author), Shanu Gupta (Author). SK Kataria & Sons.
3. Tobias Zwingmann , AI-Powered Business Intelligence: Improving Forecasts and Decision Making with Machine Learning (Grayscale Indian Edition), June 2022, Shroff/O'Reilly.

CO PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3		2	2				2
CO2	3	2	2	3				2
CO3	3		2	2				2
CO4	3		2	3				2
CO5	3		2	2				2

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3432	BIG DATA ANALYTICS	45	0	0	45	90	3

LEARNING OUTCOMES

CO1	Describe the fundamental concepts of Big Data, its characteristics, challenges, and its impact on business transformation.
CO2	Utilize Hadoop ecosystem tools and frameworks to collect, process, and manage Big Data efficiently.
CO3	Apply data analytics techniques to clean, transform, and analyze Big Data for insightful decision-making.
CO4	Evaluate real-world applications of Big Data across industries and propose innovative solutions for business improvements.
CO5	Assess security and privacy concerns in Big Data and recommend appropriate governance frameworks and protective measures.

UNIT 1 INTRODUCTION TO BIG DATA **8**

Introduction to Big Data Analytics – Types of Data - Characteristics of Big Data – Traditional vs Big Data Business Approach - Big Data and Business Transformation – Challenges in Big Data.

UNIT 2 HADOOP ECOSYSTEM **10**

Techniques and Procedures for Data Gathering, Processing, and Analysis - Big Data Platforms - Hadoop Architecture - HDFS and MapReduce - HIVE and DRILL - MAHOUT and SPARK MLlib - PIG and HIVE Framework - HBASE, ZOOKEEPER and AMBARI - SPARK - KAFKA, and STORM - OOZIE and YARN - Flume and Sqoop - Streaming Data.

UNIT 3 BIG DATA ANALYTICS **9**

Types of Big Data Problems - Data Analytics Life Cycle – Data Cleaning – Data Transformation – Descriptive Analytics – Predictive Analytics – Data Mining - Real-Time Stream Analysis - Analytic Techniques and Tools used in Big Data Visualization – Big Data Architectures – Lambda and Kappa.

UNIT 4 BIG DATA APPLICATIONS **10**

Big Data Use Cases in Various Domains - Applications in Marketing, Finance, Operations, and Human Resources - Proof of Concept (POC) - Testing and Evaluation - Social Media and Web Analytics - Sentiment Analysis - Stock Market Predictions - Text Mining - Industry-Specific Big Data Applications.

UNIT 5 SECURITY AND PRIVACY FOR BIG DATA ANALYTICS **8**

Security and Privacy Challenges in Big Data - Data Sharing and Privacy Concerns - Privacy-Preserving Methods - Authentication and Encryption Protocols - Access Control Mechanisms - Data Usage Agreement Policies - Big Data Governance Framework.

TOTAL 45 HOURS

REFERENCE BOOKS

1. Nathan Marz & James Warren, Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Manning Publications.
2. Tom White, Hadoop: The Definitive Guide, O'Reilly Media, 4th Edition.
3. Foster Provost & Tom Fawcett, Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking, O'Reilly Media.

CO PO MAPPING

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	3	3	3	3			2
CO2	3	3	2	3	3			2
CO3	3	3	2	2	2	2		2
CO4	3	3	3	3	3	2		2
CO5	3	3	3	3	3	3	3	3

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
PBA3433	BLOCK CHAIN TECHNOLOGY FOR BUSINESS DECISIONS	45	0	0	45	90	3

LEARNING OUTCOMES

CO1	To get an idea about the history of Blockchain.
CO2	Understand key features, different types of platforms & languages of blockchain technology.
CO3	Students will be familiar with cryptocurrency concepts.
CO4	Understand the design principles of ethereum.
CO5	Learn about hyperledger fabric models and its architecture.

UNIT 1 INTRODUCTION **9**

Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof.

UNIT -2 BLOCKCHAIN & APPLICATIONS **9**

Introduction to Blockchain, Gartner's Hype Curve and Evolution of Blockchain Technology, Blockchain Need & Genesis, Key Characteristics of Blockchain, Blockchain Structure, Blockchain types and Network, Mining and Consensus, How Blockchain Works, Bitcoin Whitepaper, Understanding Bitcoin, Components of a Block, Forks: soft & hard forks, Ummer blocks, Different forks from Bitcoin, Wallets, Transactions, Public & Private keys, Blockchain Applications : Internet of Things, Medical Record Management System, Do- main Name Service and future of Blockchain.

UNIT 3 CRYPTOCURRENCY **9**

History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin. Cryptocurrency Regulation: Stakeholders, Roots of Bitcoin, Legal Aspects – Cryptocurrency Exchange, Black Market and Global Economy.

UNIT 4 ETHEREU **9**

Need of Ethereum, Ethereum Foundation, Ethereum Whitepaper, How Ethereum Works, Ethereum network, Ethereum Virtual Machine, Transactions and Types, Mining & Consensus, Smart Contracts.

UNIT 5 HYPERLEDGER FABRIC **9**

Hyperledger, Hyperledger Fabric, Comparison between Fabric & Other Technologies, Fabric Architecture, Components of Hyperledger Fabric, Advantages of Hyperledger Fabric Blockchain, How Hyperledger Fabric Works.

TOTAL 45 HOURS

REFERENCE BOOKS

1. Imran Bashir, Mastering Blockchain, Packt Publishing, March 2017.
2. Debajani Mohanty, Block Chain : From Concept to Execution, BPB Publications, 2nd edition, 2018.
3. Artemis Caro, Blockchain: Bitcoin, Ethereum & Blockchain: The Beginners Guide to Understanding the Technology Behind Bitcoin & Cryptocurrency, 2017.
4. Andreas M. Antonopoulos, Gavin Wood, Mastering Ethereum: Building Smart Contracts and DApps, O'REILLY, 2018.

CO PO MAPPING

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	2	2	3			1	3
CO2	3	1	2				1	3
CO3	3	2	3				1	3
CO4	3	1	2	2			1	3
CO5	3	2	2	2			1	3